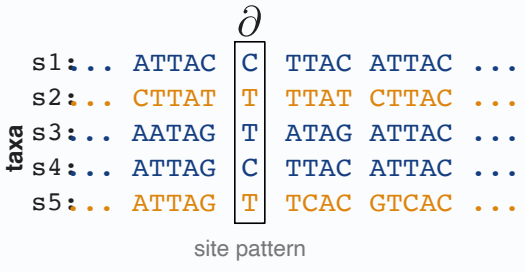
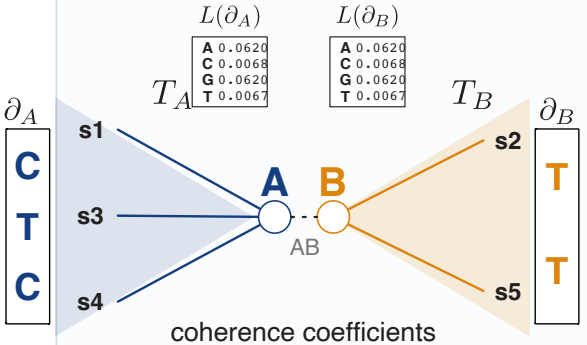
# SatuTe in a nutshell

ML-tree inference

## a Alignment and site pattern




## b Branch likelihood at AB




data re-evaluation

SatuTe

## c Weighted coherence and z-score

branch-length weights

$$w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$$

site-pattern coefficient

$$C^{\lambda}_{\partial}(t) = \sum_{i=1}^{d} w_i(t) C_{\partial i}$$

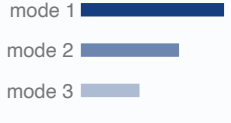
alignment average

$$\widehat{C}_{\lambda}(t) = \sum_{\partial} \frac{n_{\partial}}{n} C^{\lambda}_{\partial}(t)$$

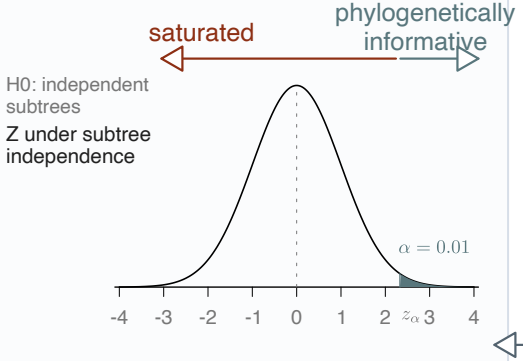
$$Z_{\lambda}(t) = \frac{\sqrt{n}\,\widehat{C}_{\lambda}(t)}{\widehat{\sigma}_{\lambda}(t)}$$

$$H_0 : \widehat{C}_{\lambda}(t) \sim \mathcal{N}(0, \hat{\sigma}^2_{\lambda}(t)/n)$$

relative persistence over AB




## d Decision under the null




reject independence when $Z_{\lambda}(t) > z_{\alpha}$